PRESIDENCY UNIVERSITY
Private University Estd. in Karnataka State by Act No. 41 of 2013
Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064

"COMMUNITY HEALTHWORKER ASSISTANCE PLATFORM"

A PROJECT REPORT

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BONAFIDE CERTIFICATE

This is to certify that, this report **"community healthworker assistance platform"** is a bonafide work of **K L sridharachari -20231CSE0167-computer science & engineering, Gabburi Navya -20231CSE0016- computer science & engineering, Thunga Hasini -20231CSE0030- computer science & engineering,** who have successfully carried out the project work and submitted the report for partial fulfilment of the requirements for the award of the degree during the academic year 2026-27**.**

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DECLARATION

We the students of final year B.Tech at Presidency University, Bengaluru, named **K L SRIDHARACHARI (20231CSE0167)-COMPUTER SCIENCE & ENGINEERING, GABBURI NAVYA (20231CSE0016)- COMPUTER SCIENCE & ENGINEERING, and THUNGA HASINI (2031CSE0030)- COMPUTER SCIENCE & ENGINEERING**, hereby declare that the project work titled "community healthworker assistance platform**"** has been independently carried out by us and submitted in partial fulfillment for the award of the degree during the academic year of 2026-27. Further, the matter embodied in the project has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.

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ABSTRACT

Community health workers frequently operate in field environments where internet connectivity may be limited or unreliable. Managing patient information, follow-up dates, vaccinations, medicines, and household visits through paper records or scattered information can make routine healthcare activities difficult to organize. This mini project presents CareConnect, a Community Health Worker Assistance Platform designed to support field-ready patient management through an Android application. The proposed platform combines digital patient registration, health records, reminders, reports, an AI chatbot, a patient field map, offline data entry, priority-based follow-up, and regional language and voice interaction.

The system is organized around a mobile application, Firebase services, local/offline storage, notification services, mapping functionality, and an AI chatbot interface. Patient information can be recorded locally when connectivity is unavailable and synchronized with Firebase when a connection is restored. A priority-based workflow is intended to organize patients according to follow-up urgency, while field mapping helps workers understand the geographic distribution of patients and households. English/Kannada language and voice interaction are proposed to improve usability in local communities. The platform is aligned with SDG 3: Good Health and Well-being by supporting continuity of care, preventive-care reminders, health information access, and organized digital records.

At Review 3, overall project completion is 80%, as reported by the project team. Earlier review material records implementation of the main interface, patient storage and priority-follow-up modules. The remaining work concerns integration, testing and final improvements. This report presents the system design, development plan, expected outputs and evaluation framework without treating mockups or planned tests as measured experimental results.

Keywords: Community health workers; Android; Room; Firebase; offline-first; priority follow-up; field map; English/Kannada; Review 3.

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ABBREVIATIONS

Abbreviation	Full form
AI	Artificial Intelligence
API	Application Programming Interface
CHW	Community Health Worker
GIS	Geographic Information System
GPS	Global Positioning System
ID	Identifier
SDK	Software Development Kit
SDG	Sustainable Development Goal
SQL	Structured Query Language
UI	User Interface
WHO	World Health Organization
XML	Extensible Markup Language

CHAPTER 1

INTRODUCTION

1.1 Background and motivation

Community health workers play an important role in delivering and supporting healthcare activities within local communities. Their work may include maintaining patient information, monitoring follow-up activities, supporting vaccination schedules, recording medicine-related reminders, and providing access to general health information. When these activities depend on paper records or scattered information, it can become difficult to organize many patients and remember important dates.

The project motivation is based on the practical needs of field workers. The project presentation identifies unreliable internet connectivity, difficulty in organizing follow-ups, and the need for quick access to general health information as important concerns. A digital platform can bring these activities into a single application while also providing mechanisms for offline operation and later synchronization.

CareConnect is proposed as a community health worker assistance platform designed for field use. The platform includes patient registration, digital health records, reminders, reports, an AI chatbot, field mapping, offline mode, priority-based follow-up, and regional language support. The design is organized into four broad layers: Presentation, Business Logic, Data/Service, and Device/Network.

A central characteristic of the platform is offline-first operation. Patient information should remain available to the health worker even when network connectivity is lost. Data entered during an offline period can be stored locally and synchronized with Firebase when the internet connection becomes available. This workflow is intended to reduce dependence on continuous connectivity during field visits.

The system also includes a field map for geographically organizing patients or households, a priority-based patient list for follow-up activities, and language/voice interaction to improve accessibility. These features are intended to make the application practical for community-based healthcare work.

CareConnect is the Community Health Worker Assistance Platform developed under project PRJ-510. This report consolidates the design, supplied application mockups, implementation description and current Review 3 progress. Overall completion is 80%, as reported by the project team for Review 3; Review 2 recorded an earlier milestone of 60%.

1.2 Problem statement

Community Health Workers need a simple digital platform to manage patient information, track healthcare activities, and access general medical information. Managing many patients and follow-up dates can become difficult with paper records or scattered information. Field workers may have limited or unreliable internet connectivity. Important follow-ups, vaccinations, medicines, and visits can be difficult to organize, while workers may need quick access to general health information during community visits.

The proposed platform addresses these needs through a unified Android application that combines patient management, reminders, reports, AI-assisted information access, field mapping, offline operation, priority-based follow-up, and regional language support.

1.3 Scope and limitations

The project scope is an Android application for community health workers, supported by Room, Firebase Authentication, Firebase Firestore, notification services, mapping and speech interaction. Its core responsibility is to organize patient records and field activities. It does not establish diagnostic accuracy, recommend individualized treatment or replace professional medical judgment.

The supplied materials describe both expected workflows and implementation progress. The available images are explicitly presented as design mockups in the architecture document. They are retained as interface illustrations rather than proof of completed runtime tests. No measured performance dataset, clinical trial, test execution log or final application package was supplied.

1.4 Methodology overview

CareConnect is proposed as a software-based Android platform. The main workflow begins with authentication and access to a dashboard. A health worker can register patients, view patient details and digital health records, manage reminders, access the patient map, and use the AI chatbot. When the device is offline, local storage supports continued data entry; when connectivity returns, the stored records can be synchronized with Firebase.

1.5 Organization of the report

Chapter 2 reviews related work and technical foundations. Chapter 3 identifies gaps motivating the project. Chapter 4 explains the proposed methodology, and Chapter 5 defines the objectives. Chapter 6 covers design and implementation. Chapter 7 presents the ten-week schedule. Chapters 8 and 9 discuss outcomes, current progress and evaluation. The report concludes with future work, references, pseudocode, interface illustrations and enclosure details.

CHAPTER 2

LITERATURE SURVEY

2.1 Community health worker support

The World Health Organization guide on integrating community health workers into health systems provides policy context for supporting community-based workers (World Health Organization, 2025a). CareConnect applies this context at a project level by organizing records, visits and follow-up activities in a mobile workflow. The guide is contextual evidence; it is not an evaluation of CareConnect.

2.2 Digital health adoption

The WHO European Region policy brief addresses uptake of digital solutions by the health and care workforce (World Health Organization Regional Office for Europe, 2025). This supports considering usability, training and implementation conditions alongside application features. For CareConnect, accessible navigation, local-language interaction and continuity during network loss are design priorities.

2.3 Offline-first mobile architecture

Android Developers describes an offline-first data layer with a local data source and a network data source (Google, n.d.-a). Local reads allow the user interface to remain useful without reliable connectivity. CareConnect adopts Room as its local record store and Firestore as the synchronized cloud store. This arrangement requires an explicit synchronization design between the two stores.

Room provides a persistence abstraction over SQLite and supports structured database access (Google, n.d.-b). The project uses it for patient records needed during field visits. Its role is different from the cloud service: Room supports device-local operations, while the cloud copy supports connected access and synchronization.

2.4 Authentication and cloud synchronization

Firebase Authentication supplies user authentication services (Google, n.d.-c). Cloud Firestore documentation describes offline access and synchronization of its own cached data (Google, n.d.-d). Firestore caching alone does not implement synchronization with a separate Room database; the application must coordinate that path and handle pending changes, retries and conflicts.

2.5 Mapping and AI information support

The Maps SDK for Android provides mapping functionality suitable for geographic patient or household presentation (Google, n.d.-e). CareConnect uses a field-map concept to support visit planning; the project does not establish route optimization or guaranteed offline map availability.

Firebase AI Logic documentation describes access to Gemini models through its client SDKs (Google, n.d.-f). The supplied paper cites this technology as a possible AI integration resource. CareConnect specifies an AI chatbot API, but the uploaded materials do not confirm a final model, model version or evaluated chatbot configuration.

2.6 Synthesis and project relevance

Table 2.1 literature and technology synthesis is presented below.

Table 2.1 Literature and technology synthesis

Source or topic	Contribution to the design	Remaining project question
WHO CHW guide (2025a)	Context for community-worker support	How will workers use the app in actual visits?
WHO digital uptake brief (2025)	Attention to adoption and usability	Are navigation and language choices understandable?
Android offline-first guidance	Local and network data separation	Can interrupted changes synchronize without loss?
Room and Firebase documentation	Local records, identity and cloud storage	Are ownership and conflict rules enforced?
Maps and AI service documentation	Geographic display and information access	Do integrations work under field conditions?

Review 2 also mentions Digital Lifecare and Padayon as examples of offline-oriented community-health systems, and identifies interoperability and duplicated data entry as concerns. Full bibliographic records were not supplied for those examples, so they are treated here as project research leads rather than verified comparative studies. The incomplete author/year entries shown in Review 1 are not used to assert specific published findings.

CHAPTER 3

RESEARCH GAPS OF EXISTING METHODS

3.1 Fragmented patient information

Separate patient records, reminder lists and household details require repeated searches or data entry. CareConnect links these workflows through a shared patient record.

3.2 Dependence on continuous connectivity

Local saves allow field data entry to continue without a cloud response. Synchronization is deferred until connectivity returns; pending changes must remain visible and recoverable after interruptions.

3.3 Follow-up organization and accessibility

High Priority, Due Today and Upcoming categories distinguish follow-up needs. English/Kannada and voice interaction address accessibility. Both task organization and language usability require evaluation with intended users.

3.4 Gap-to-feature mapping

Table 3.1 field problems and CareConnect responses is presented below.

Table 3.1 Field problems and CareConnect responses

Identified Problem	Proposed Platform Response
Paper/scattered patient information	Digital patient records
Unreliable internet	Offline mode + synchronization
Difficult follow-up organization	Priority categories + reminders
Geographically distributed households	Patient field map
Need for general health information	AI chatbot
Local language / interaction needs	English/Kannada + voice support

3.5 Boundary of the contribution

The contribution is an integrated field-workflow design. The supplied evidence does not establish novelty over all existing products, comparative superiority, national-system interoperability, clinical validation or deployment at scale.

CHAPTER 4

PROPOSED METHODOLOGY

4.1 Development approach

The methodology progresses from problem identification and requirement analysis to layered design, Android interface development, local storage, service integration and testing. Features are organized around one field workflow: identify the patient, record the visit, review follow-up needs, and retain the information even when the device is offline.

Health worker logs into the Android application.

Worker accesses the dashboard and patient-management functions.

New patient information is registered or existing patient records are opened.

Records are stored locally when offline and synchronized with Firebase when connectivity returns.

Patient information is evaluated through priority rules for follow-up organization.

Reminders support medicines and vaccinations.

Field-map functionality displays patients/households geographically.

AI chatbot and voice/language functions provide information-access support.

4.2 Authentication and record entry

The application begins with login and authentication. Firebase Authentication is identified in the technology stack as the service for managing authentication. The purpose is to provide controlled access to the application's patient-management functions.

The patient-registration module allows a community health worker to create a digital patient record. Instead of depending on paper records, patient information is organized within the mobile application. The patient list and patient-details screens provide access to stored records.

Digital health records provide a structured way to organize information associated with patients. The project proposes maintaining records so that workers can access relevant information during field activities. Local storage is particularly important when network access is unavailable.

4.3 Offline data entry and synchronization

The offline workflow is one of the central features of the proposed platform. The intended sequence is: Patient data > Room > offline use > internet returns > Firebase sync. This enables a worker to continue recording information without waiting for network availability.

Patient Data > Room Database > Offline Use > Connectivity Restored > Firebase Sync

Figure 4.1 reproduces the supplied offline workflow. The design intent is to write locally first in both online and offline conditions, then synchronize pending records. This avoids making a successful local save depend on an immediate network response.

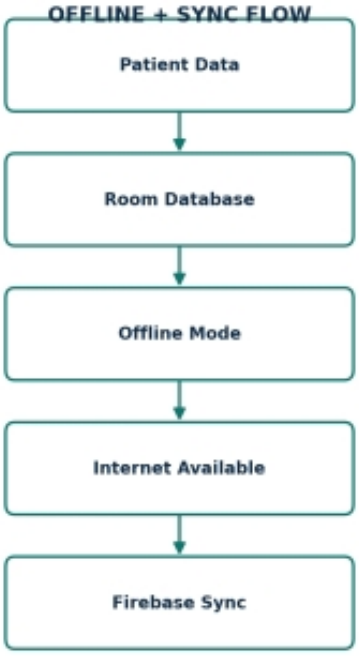


Figure 4.1 Offline storage and synchronization workflow

Additional design detail: each local change should carry a stable record identifier and synchronization state. A queued change should be removed only after a confirmed remote acknowledgement. If the local record changes during an upload, the newer change must remain pending. These are proposed reliability rules, not a claim that the current build has passed concurrency testing.

4.4 Priority-based follow-up

The priority workflow is designed to organize patients according to follow-up urgency. The project identifies categories such as High, Due Today, and Upcoming. The workflow is: Patient data > priority rules > High / Due Today / Upcoming > worker action.

Patient Data > Priority Rules > High / Due Today / Upcoming > Follow-up Action

Figure 4.2 shows the supplied priority workflow. The exact priority thresholds were not provided. The illustrative rules in Appendix A therefore use follow-up dates and a worker-reviewed flag, rather than presenting an unvalidated medical risk score.

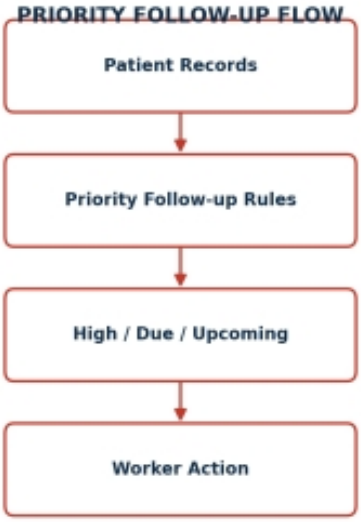


Figure 4.2 Priority follow-up workflow

4.5 Regional language and voice

The project proposes English and Kannada language support together with voice interaction. The intended workflow is: English/Kannada > voice input > speech-to-text > record/assistant. This feature is designed to make the application easier to use in local community settings.

Figure 4.3 shows how speech input becomes text for a record or assistant query. The worker should review the transcript before saving. Text entry should remain available when speech recognition is unavailable or inaccurate.

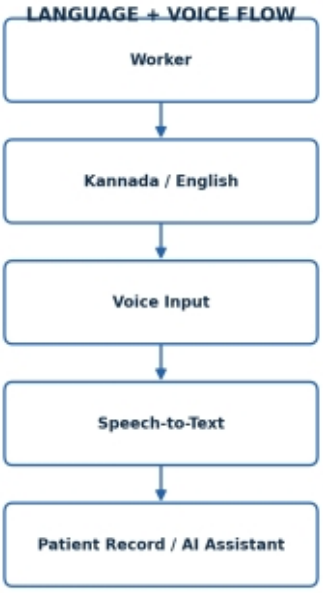


Figure 4.3 Language and voice workflow

4.6 Field mapping and reminders

The patient field map enables a health worker to see patients or households in the assigned area on a map. This is intended to help organize geographically distributed patients and reduce the need to remember household locations manually. GIS or field-map functionality is included in the software requirements.

Android notification services support medicine and vaccination reminders. These reminders are intended to help workers organize routine follow-up activities and reduce the possibility of missing scheduled tasks.

Map markers should open the matching patient details so that location remains linked to the correct record. Reminders should store their scheduled time, patient identifier and completion state. Changes to a follow-up date should update the corresponding reminder instead of leaving a stale notification.

4.7 Data preparation and validation

CareConnect is a transactional patient-management application; the supplied project does not describe a machine-learning training dataset. Input preparation therefore means validating entered records, checking required fields, using consistent dates, linking follow-ups to patients and allowing corrections before saving. Demonstrations and tests should use fictitious records rather than unnecessary personal patient data.

CHAPTER 5

OBJECTIVES

5.1 Project objectives

CareConnect aims to: (1) provide Android-based patient registration and digital records; (2) enable Room-based offline access and Firebase synchronization; (3) organize priority follow-up and medicine/vaccination reminders; (4) provide patient/household mapping and general-information chatbot access; and (5) support English/Kannada and voice interaction. These functions support the project's SDG 3 theme.

5.2 Objective verification

Table 5.1 objectives and proposed acceptance evidence is presented below.

Table 5.1 Objectives and proposed acceptance evidence

Objective	Evidence required for completion
Digital patient management	Create, retrieve, edit and validate a fictitious patient record.
Offline-first operation	Save in airplane mode, restart the app and retrieve the same record.
Cloud synchronization	Reconnect and compare local and remote versions without duplicate records.
Priority and reminders	Check overdue, due-today and future cases; observe reminder behavior.
Field map	Open a marker and confirm its linked patient record.
English/Kannada and voice	Check translated labels and review a captured transcript.
AI information support	Check response, failure handling and clear information-only labeling.

5.3 SDG alignment

CareConnect aligns with Sustainable Development Goal 3 through organized records, preventive-care reminders and continuity of field work (United Nations, n.d.). Appendix C gives the feature-level mapping; health impact has not been measured.

CHAPTER 6

SYSTEM DESIGN AND IMPLEMENTATION

6.1 System architecture

The platform is organized into four conceptual layers: Presentation, Business Logic, Data/Service, and Device/Network. This separation helps organize the application into user-interface functions, application rules, data management, and device/network capabilities.

Figure 6.1 reproduces the architecture supplied with the project. It separates interface functions, application rules, storage/services and device/network capabilities.

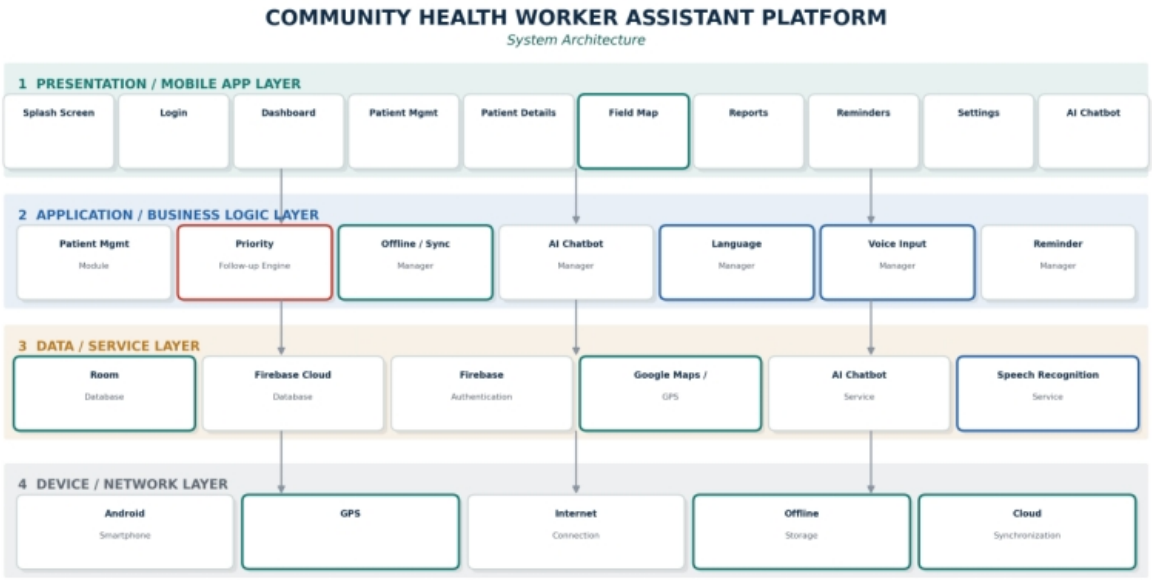


Figure 6.1 Four-layer CareConnect architecture

Table 6.1 responsibilities of the four architectural layers is presented below.

Table 6.1 Responsibilities of the four architectural layers

Layer	Major Components	Purpose
Presentation	Android UI, Dashboard, Patient screens, Settings	Provides the interface used by the health worker.
Business Logic	Patient management, Priority rules, Reminders, Workflows	Processes application operations and follow-up logic.

Table 6.1 Responsibilities of the four architectural layers

Layer	Major Components	Purpose
Data / Service	Room/local records, Firebase Authentication, Firestore, AI Chatbot API	Stores patient information and connects the application to cloud/AI services.
Device / Network	Android device, Notifications, GPS/Map, Internet connectivity	Provides field-device capabilities and handles online/offline conditions.

6.1.1 Presentation Layer

The presentation layer contains the Android user interface. The project identifies login and authentication, dashboard, patient registration, patient list and details, digital health records, AI chatbot, medicine reminders, vaccination reminders, and profile/settings as important application screens and functions.

6.1.2 Business Logic Layer

The business logic layer coordinates the application's core workflows. It includes patient management, follow-up priority classification, reminders, and application actions. Priority rules organize patient follow-up into categories such as High, Due Today, and Upcoming.

6.1.3 Data and Service Layer

The data/service layer includes local patient records and cloud services. The project specifies Room database for local/offline patient records and Firebase Firestore for synchronized cloud data. Firebase Authentication is identified for authentication. The AI chatbot API provides the proposed AI-assisted information function.

6.1.4 Device and Network Layer

The device/network layer supports Android smartphone or emulator use, notifications, mapping/GIS functions, and network connectivity. Since connectivity can be unreliable in field conditions, the architecture explicitly supports offline operation and later synchronization.

6.2 Software and hardware requirements

Table 6.2 development and runtime requirements is presented below.

Table 6.2 Development and runtime requirements

Component	Technology or requirement	Purpose
Development	Android Studio; Java/XML; Kotlin noted in Review 2	Application screens and logic
Local persistence	Room database	Patient access and offline records
Cloud services	Firebase Authentication and Firestore	Identity and synchronized data
Field support	Maps/GIS and location services	Patient or household display
Interaction	English/Kannada; speech-to-text	Regional-language and voice workflows
Other services	Android notifications; AI chatbot API	Reminders and information access
Development hardware	Laptop or desktop computer	Build, debug and document the application
Test hardware	Android smartphone or emulator	Functional and integration testing

No minimum Android version, RAM, storage capacity, device model, service tier or final dependency versions are specified in the supplied files. These values should be recorded from the actual Review 3 build rather than guessed.

6.3 Module-wise design

The implementation is planned as an Android application using Java and XML. Firebase services provide cloud-based authentication and synchronized storage, while Room provides local storage for offline operation. Android notification services are used for reminder-related workflows. The AI chatbot API is included as a service component for general health-information assistance.

6.3.1 Android Application

The Android application acts as the primary interface for community health workers. It provides the dashboard, patient screens, records, reminders, map, chatbot, profile, and settings. The application is designed around field usability, with important operations accessible from the mobile device.

6.3.2 Local Storage

Room database is identified as the local/offline storage mechanism. Patient records can be maintained locally so that data entry does not depend on an active network connection. Local data becomes part of the synchronization workflow when internet connectivity returns.

6.3.3 Firebase Services

Firebase Authentication is used for user authentication, while Firebase Firestore is identified for cloud-synchronized data. The cloud layer provides a persistent service for records that have been synchronized from the field device.

6.3.4 Notification Services

Android notification services support medicine and vaccination reminders. These reminders are intended to help workers organize routine follow-up activities and reduce the possibility of missing scheduled tasks.

6.3.5 AI Chatbot API

The AI chatbot is proposed as an application feature for general health information. The chatbot can be accessed from the Android application through an AI Chatbot API. The design should keep the distinction between general information and individualized medical advice clear.

6.3.6 Mapping / GIS

GIS or map functionality supports the patient field map. Patients or households can be represented geographically so that workers can understand the spatial distribution of their assigned community.

6.3.7 Voice and Language Support

Speech-to-text and regional language support are proposed to make interaction easier in local community settings. English and Kannada are identified as supported languages in the project requirements.

The reporting module is intended to summarize operational activities such as patient counts and follow-up status. The settings screen centralizes profile, language and application options. Counts visible in the supplied mockups are sample interface values and are not project evaluation measurements.

6.4 Suggested logical data model

Table 6.3 adds a logical schema consistent with the supplied workflows. It is a proposed implementation aid; it is not extracted from a submitted database or repository.

Table 6.3 suggested entities and relationships is presented below.

Table 6.3 Suggested entities and relationships

Entity	Suggested fields	Relationship or rule
Worker	workerId, displayName, language	Owns or is assigned authorized records.
Patient	patientId, workerId, name, localVersion	Stable identifier; linked to worker.
Visit	visitId, patientId, visitDate, notes	Multiple visits can belong to one patient.
FollowUp	followUpId, patientId, dueDate, status, flag	Linked to patient; determines task category.
Reminder	reminderId, patientId, type, scheduledAt	Medicine or vaccination reminder.
Location	patientId, latitude, longitude, accuracy	Optional; coordinates require validation.
SyncOperation	operationId, entityId, version, state	Pending change retained until acknowledgement.

6.5 Component integration

The user interface should obtain patient data through a repository that coordinates local storage and service requests. A local transaction records a patient change and the associated pending operation together. The follow-up list uses stored dates and status, while the map uses the same patient identifier to open details. This avoids separate copies of patient identity across features.

The synchronization component needs retry handling, stable identifiers and version checks. After a remote acknowledgement, it should mark only the acknowledged local version as synchronized. Remote changes that conflict with unsent local edits should be retained for reconciliation rather than silently replacing the local record. Appendix A supplies illustrative pseudocode for these operations.

6.6 Security and responsible information handling

Patient information is sensitive application data and should be handled carefully. Firebase Authentication is included in the proposed stack to control user access. Cloud synchronization should be designed so that records are transferred only through the intended authenticated application workflow. Local storage should also be protected according to the capabilities of the Android platform.

The AI chatbot should not be treated as an independent medical decision-maker. Its purpose in this project is general health-information support. The application should distinguish between stored patient information, general information, and AI-generated responses so that users understand the nature of the information being presented.

A community health application must prioritize responsible handling of patient information. The system should limit access to authorized users and avoid exposing personal information unnecessarily. Data synchronization should be performed securely, and the application should clearly distinguish recorded patient information from AI-generated content.

The AI chatbot should provide general health information rather than present itself as a medical professional. For consequential healthcare decisions, users should be encouraged to consult qualified healthcare professionals. The project therefore positions AI as an assistive component within the broader platform.

Additional design checks include restricting cloud records to authorized users, testing access denial for another account, limiting sensitive information in notifications and logs, and keeping service credentials out of published source code. Authentication alone is not evidence that every database access is properly authorized.

6.7 Individual contributions

Table 6.4 team contributions recorded in Review 2 is presented below.

Table 6.4 Team contributions recorded in Review 2

Team member	Recorded contribution
K L Sridharachari 20231CSE0167	Android application structure, Java/XML development, integration of modules and Firebase/services, testing and overall integration.
Gabburi Navya 20231CSE0016	Patient registration and records, local storage, offline-first handling, user interface, testing and documentation.
Thunga Hasini 20231CSE0030	Reminders and follow-ups, chatbot and map functionality, language/voice features, testing, documentation and integration.

These are the supplied contribution descriptions. The Review 3 update changes overall progress to 80%; no revised person-by-person allocation was supplied. The source materials identify the repository as Sridhark-09/community-health-worker-assistance-platform. No repository source snapshot is included in the uploaded files.

CHAPTER 7

TIMELINE FOR EXECUTION OF PROJECT

7.1 Ten-week development plan

The project presentation defines a ten-week development timeline. The activities cover problem identification, literature survey, requirements, system design, UI development, Firebase integration, patient management, AI chatbot development, reminders/reports, testing, and documentation.

Figure 7.1 presents the original ten-week activity plan from the project paper. The bars show planned activity windows, not measured percentage completion. Calendar dates are not inferred from week numbers.

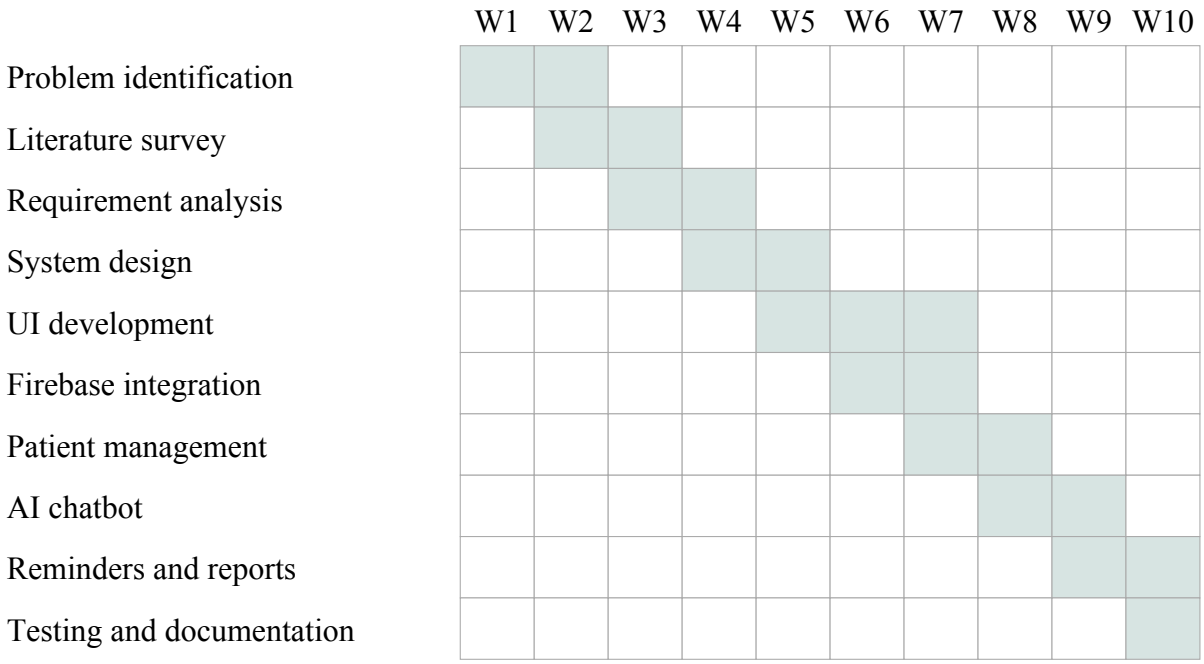


Figure 7.1 Planned ten-week development schedule

7.2 Development phases

Problem identification: define the field problems faced by community health workers.

Literature survey: review community health worker, digital health, and related technology sources.

Requirement analysis: identify functional and technology requirements.

System design: define application layers, workflows, and data/service components.

UI development: develop the Android screens and navigation.

Firebase integration: connect authentication and cloud data services.

Patient management: implement patient registration, lists, details, and records.

AI chatbot: integrate the chatbot API for general health-information access.

Reminders and reports: implement medicine/vaccination reminders and reporting functions.

Testing and documentation: test workflows and prepare project documentation.

7.3 Review 3 milestone and remaining work

The project team reports 80% completion for Review 3. This supersedes the 60% milestone recorded in Review 2. The update is an overall progress estimate; it does not provide a new breakdown of completed integration tests or feature-by-feature acceptance results.

The remaining work is organized around final integration, verification and improvements: reconcile offline and cloud data under interruptions, check map and voice/language behavior, validate reminders and reporting, complete end-to-end testing, and assemble final demonstration evidence. These are completion tasks based on the last documented open areas and should be checked against the current build.

CHAPTER 8

OUTCOMES

8.1 Expected application outputs

The project presentation identifies a set of expected application functions and outputs. The final Android application is expected to provide an integrated experience for community health workers rather than separate tools for individual tasks.

The expected outputs include login, dashboard, registration, patient lists and details, digital health records, chatbot interaction, medicine/vaccination reminders, a field map, priority lists, offline records, Firebase synchronization, English/Kannada and voice interaction, and profile/settings. Table 8.1 explains the function of each feature.

8.2 Expected user experience

A health worker should be able to log into the platform, quickly locate a patient, create or update a record, check follow-up priority, and use reminders or map functions as needed. The application should continue to support important record operations when connectivity is unavailable. Once connectivity returns, synchronization should reduce the need for manual transfer of records.

The dashboard is intended to provide a central entry point to the main functions. Patient management should remain straightforward, while the priority list and map should help the worker decide which field activities require attention. The AI chatbot is intended to provide quick access to general health information, and regional language/voice support is intended to reduce interaction barriers.

8.3 Reported implementation outcomes

Review 2 reports implemented screens/modules for Splash, Login, Dashboard, Patient Management, Patient Details, Offline Mode, Priority Follow-up, Field Map, Language/Voice, Reports and Settings. It also reports patient-data storage and priority-follow-up output. The Review 3 update increases overall completion to 80%.

The earlier review says core modules were tested and that Firebase synchronization, map, voice/language and final integration were under testing. The current progress update does not specify which of these tests are now complete. Accordingly, this report records the implemented-module claims and preserves the outstanding verification boundary.

8.4 Feature coverage

Table 8.1 careConnect functions and intended outputs is presented below.

Table 8.1 CareConnect functions and intended outputs

Feature	Description
Login and Authentication	Provides controlled access to the application.
Dashboard	Provides a central view of the application's major functions.
Patient Registration	Allows health workers to create digital patient records.
Patient List and Details	Supports viewing and organizing patient information.
Digital Health Records	Maintains structured patient-related information.
AI Chatbot	Provides access to general health-information support through an AI API.
Medicine Reminders	Supports routine medicine-related reminders.
Vaccination Reminders	Supports preventive-care reminder activities.
Patient Field Map	Displays patients/households geographically in the assigned area.
Priority Follow-up	Organizes patients according to follow-up priority.
Offline Mode	Allows patient data entry without continuous connectivity.
Firebase Synchronization	Synchronizes local records with cloud data when connectivity returns.
Regional Language / Voice	Supports English/Kannada and voice interaction.
Profile and Settings	Provides user/application configuration options.

8.5 Practical value and limits

The intended outcome is a single workspace for field records, follow-up organization and information access. Local persistence can reduce dependence on a live network for record entry. Priority lists and maps can make planned field work easier to organize. These are expected benefits of the design; reduced workload, improved adherence or better health outcomes have not been measured in the supplied materials.

CHAPTER 9

RESULTS AND DISCUSSIONS

9.1 Current Review 3 status

Overall project completion is 80%, based on the team's latest Review 3 update. The earlier 60% figure belongs to Review 2. The current report is therefore an implementation-progress report rather than a claim of a fully validated production deployment.

Table 9.1 evidence and interpretation of current progress is presented below.

Table 9.1 Evidence and interpretation of current progress

Evidence	What it supports	What it does not establish
Review 3 update: 80%	Current team-reported overall progress	A measured pass rate or feature-by-feature completion.
Review 2 implementation slide	Reported screens, local storage and priority outputs	Independent confirmation of the running build.
Architecture document and mockups	Layered design and intended screen behavior	Screenshots from executed acceptance tests.
Research-paper testing plan	Intended test coverage and expected outputs	Observed outcomes or measured performance.

9.2 Testing approach and environment

The evaluation environment should use the actual Android device or emulator, the Review 3 application build, a test Firebase project and fictitious patient records. Record the Android version, app version, network state and relevant permissions for each execution. Repeat core record workflows online, offline and after application restart.

Table 9.2 contains the supplied functional test plan. Expected results require actual execution and recorded observations before being marked as passed.

Table 9.2 functional testing plan and expected results is presented below.

Table 9.2 Functional testing plan and expected results

Test Area	Test Objective	Expected Result
Authentication	Verify valid and invalid login behavior.	Authorized users can access the application.
Patient registration	Create and save a patient record.	Patient information is stored correctly.
Patient retrieval	Open patient list and details.	Correct patient information is displayed.
Offline entry	Create/update records without internet.	Records remain available locally.
Synchronization	Restore network after offline activity.	Local records synchronize with Firebase.
Priority follow-up	Apply configured priority rules.	Patients are organized into priority categories.
Medicine reminders	Schedule and trigger a medicine reminder.	Reminder notification is generated.
Vaccination reminders	Schedule vaccination follow-up.	Reminder is available to the worker.
Field map	Open map and view patient/household locations.	Relevant locations are displayed geographically.
AI chatbot	Send a general health-information query.	A response is returned through the chatbot API.
Voice input	Use supported voice interaction.	Speech is converted to text for the intended workflow.
Language support	Use English/Kannada interface or interaction.	Supported language interaction is available.

9.3 Additional integration and failure tests

Table 9.3 suggested reliability and integration tests is presented below.

Table 9.3 Suggested reliability and integration tests

Test situation	Expected acceptance condition
Offline save then restart	The same record is still readable; pending status is retained.
Network loss during upload	A retry is safe and does not create a duplicate record.

Table 9.3 Suggested reliability and integration tests

Test situation	Expected acceptance condition
Local edit while upload runs	Acknowledging an older version does not clear the newer pending edit.
Two devices edit one record	Conflict is detected or resolved under an explicit documented policy.
Permission denied for location or microphone	The app explains the limitation and keeps text/manual workflows usable.
Cloud/API service unavailable	Local records remain available; failed requests show a retryable status.
Unauthorized account requests a record	Access is denied; another worker's private records are not exposed.
Date change or time-zone change	Follow-up categories and scheduled notifications remain consistent.

9.4 Performance measures to collect

Useful measures include local save time, patient-list loading time, synchronization completion time, retry success, duplicate-record count, reminder delivery behavior and task completion time. Each measure requires a defined device, data volume, network condition and repeat count. No measured values are supplied, so this report does not state accuracy percentages, response-time reductions or clinical effectiveness.

9.5 Limitations and interpretation

No completed quantitative evaluation results were supplied. Synchronization must handle conflicts and interruptions; priority categories depend on configured rules. Mapping requires suitable location data and services. Voice and regional-language performance depend on speech-recognition availability and quality. AI responses need review, and the project does not establish clinical decision-making capability.

The 80% progress update is retained as a team-reported milestone. Detailed completion evidence, runnable source/build verification, finalized priority thresholds and measured test results remain necessary before claiming complete system acceptance.

CONCLUSION AND FUTURE WORK

Conclusion

This mini project presents CareConnect, a Community Health Worker Assistance Platform designed for field-oriented healthcare support. The proposed Android application brings patient registration, digital records, reminders, reports, AI-assisted information access, field mapping, offline operation, priority-based follow-up, and regional language/voice interaction into one platform.

The offline-first design is a central part of the project because community health workers may operate in areas with limited connectivity. The intended workflow allows records to be stored locally and synchronized with Firebase when connectivity returns. The priority list, reminders, and map provide additional support for organizing field activities.

The project also maps its features to SDG 3: Good Health and Well-being through patient tracking, vaccination reminders, health-information access, medicine reminders, and digital record organization. Future work can strengthen synchronization, priority classification, multilingual interaction, mapping, analytics, and AI safeguards.

Overall, CareConnect provides a structured foundation for a field-ready community health worker assistance platform, with emphasis on practical mobile workflows, offline availability, organized patient information, and accessible digital support.

For Review 3, the current completion level is 80%. The report documents the implemented-module descriptions available from the earlier review and the current overall progress update, while keeping expected behavior separate from measured results. Completing integration and acceptance evidence is the next step toward a final demonstration.

Future work

Enhanced Offline Synchronization

Future versions can improve synchronization handling for conflicts, interrupted uploads, and larger amounts of locally stored information. A robust synchronization mechanism can improve reliability in areas with intermittent connectivity.

Advanced Priority Classification

Priority-based follow-up can be expanded with more detailed rules and configurable categories. Future versions could support additional factors while maintaining transparent and understandable prioritization.

Expanded Regional Language Support

The current project identifies English and Kannada. Additional Indian languages can be incorporated in future versions to improve accessibility across different communities.

Improved Voice Interaction

Voice-based workflows can be expanded to support more application actions, provided that speech recognition remains accurate in the intended field environment.

Enhanced Mapping

The field map can be extended with visit planning, route support, geographic filtering, and other field-management capabilities.

Improved AI Information Support

The AI chatbot can be enhanced with stronger safeguards, carefully selected health-information sources, and clearer separation between general information and professional medical guidance.

Analytics and Reporting

Future versions can include dashboards and reports showing follow-up status, vaccination activities, medicine reminders, patient distribution, and other operational indicators.

Scalable Cloud Architecture

The Firebase-based design can be expanded as the number of users, communities, and patient records grows. Additional cloud services may be considered when required by the deployment scale.

REFERENCES

References are listed in order of first citation, using author-date citations in the report body. Online documentation was accessed on 24 October 2026. The project's own uploaded materials are identified separately below.

[1] World Health Organization (2025a). Integrating community health workers into health systems: a step-by-step policy implementation guide. Available at: https://www.who.int/publications/i/item/9789240110298

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[4] Google (n.d.-b). Save data in a local database using Room. Android Developers. Available at: https://developer.android.com/training/data-storage/room

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[8] Google (n.d.-f). Gemini API using Firebase AI Logic. Available at: https://firebase.google.com/docs/ai-logic

[9] United Nations (n.d.). Goal 3: Ensure healthy lives and promote well-being for all at all ages. Available at: https://sdgs.un.org/goals/goal3

Project source materials

Sridharachari, K. L., Navya, G. and Hasini, T. (2026). CareConnect: An Offline-First Digital Platform for Community Health Workers. Supplied mini-project research-paper manuscript, Presidency University.

Project team (2026). Review 1 and Review 2 presentations for Community Health Worker Assistance Platform, PRJ-510. Supplied PDF presentations, Presidency University.

Project team (2026). Community Health Worker Assistance Platform: System Architecture, Expected Application Output and Gantt Chart. Supplied AD&GC document.

Project team (2026). Review 3 progress update: overall completion 80%. Update supplied for preparation of this report.

APPENDIX A PSEUDOCODE

The following pseudocode is an illustrative design addition derived from the supplied workflows. It is not extracted from the project repository and is not presented as tested application source code. Names and rules should be aligned with the actual Review 3 build.

A.1 Local patient save

1. Receive the patient form and the current authorized worker identifier.

2. Validate required fields, dates and any supplied coordinates. Show field-specific errors if validation fails.

3. Begin a local transaction. Reuse the stable patient identifier for an edit; generate one for a new record.

4. Increase the local record version, save the patient and add or update the pending synchronization operation.

5. Commit both changes together. Refresh the patient list from Room and display that the record is saved locally.

6. Request synchronization when an authenticated network connection is available.

A.2 Synchronization with version acknowledgement

1. If network access or an authorized session is unavailable, retain pending operations and exit.

2. Select a pending operation and capture the entity identifier, operation identifier and local version.

3. Send the change using stable identifiers so that retrying the same operation does not create a second record.

4. If the remote version conflicts with the intended change, retain both versions and flag the record for reconciliation.

5. If the server acknowledges the operation, re-read the local version in a transaction.

6. Mark the record synchronized only if its version still equals the uploaded version. Otherwise retain the newer pending change.

7. On failure, keep the operation pending and schedule a bounded retry. Do not delete unsent changes.

8. Apply permitted remote updates without overwriting unsent local edits; record the last successful synchronization status.

A.3 Illustrative follow-up classification

These rules organize tasks by date and worker review; they are not clinical diagnosis or triage rules. Exact project priority thresholds were not supplied.

1. Read the follow-up due date, task status and worker-reviewed priority flag.

2. If the task is complete, remove it from the active pending list.

3. If the record needs urgent worker review or its follow-up is overdue, assign High Priority.

4. Otherwise, if the due date equals the current local date, assign Due Today.

5. Otherwise, if the due date is later than today, assign Upcoming.

6. If the date is missing, show Needs Scheduling rather than silently assigning a future date.

7. Sort active tasks by category and date. Explain the reason for the category in patient details.

A.4 Voice-assisted record entry

1. Read the selected English/Kannada preference and check microphone permission.

2. If permission or the speech service is unavailable, keep manual text entry active.

3. Capture speech and obtain a transcript in the selected supported language.

4. Display the transcript for worker correction and confirmation.

5. Validate the confirmed text, then send it to the intended form field or general-information assistant.

6. Do not save an unreviewed transcript directly into a patient record.

A.5 Reminder and field-map operations

1. Create or update a reminder linked to the patient and intended follow-up time.

2. Check notification permission and explain any restriction to the worker.

3. When the reminder is due, show only the information needed to open the relevant task.

4. When a task is completed or rescheduled, cancel or replace its earlier reminder.

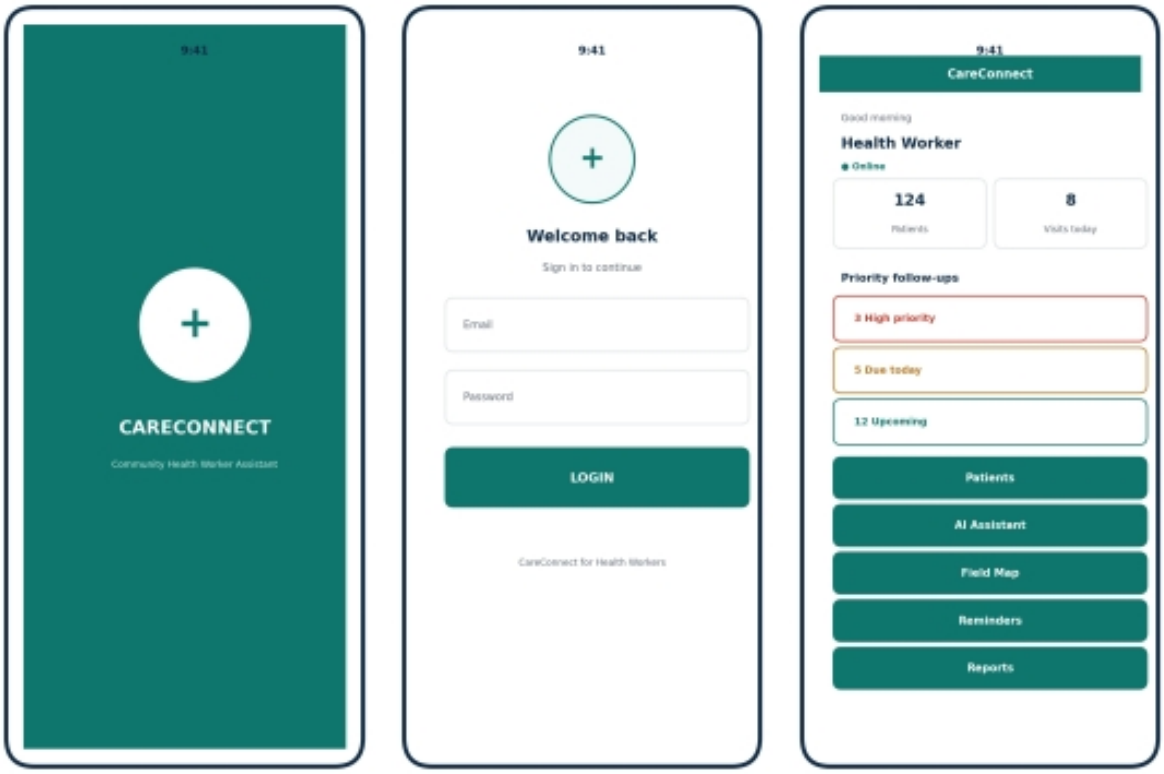
5. For the map, validate available coordinates and display markers linked to stable patient identifiers.

6. On marker selection, open the matching patient details; keep the patient list usable when the map cannot load.

APPENDIX B SCREENSHOTS AND INTERFACE MOCKUPS

The supplied AD&GC document labels these images as expected application output. They are reproduced as design mockups, not verified runtime screenshots. All twelve original screens are included. Figure B.1 covers entry screens, Figure B.2 covers patient management, Figure B.3 covers field work and Figure B.4 covers support and settings.

B.1 Entry and dashboard



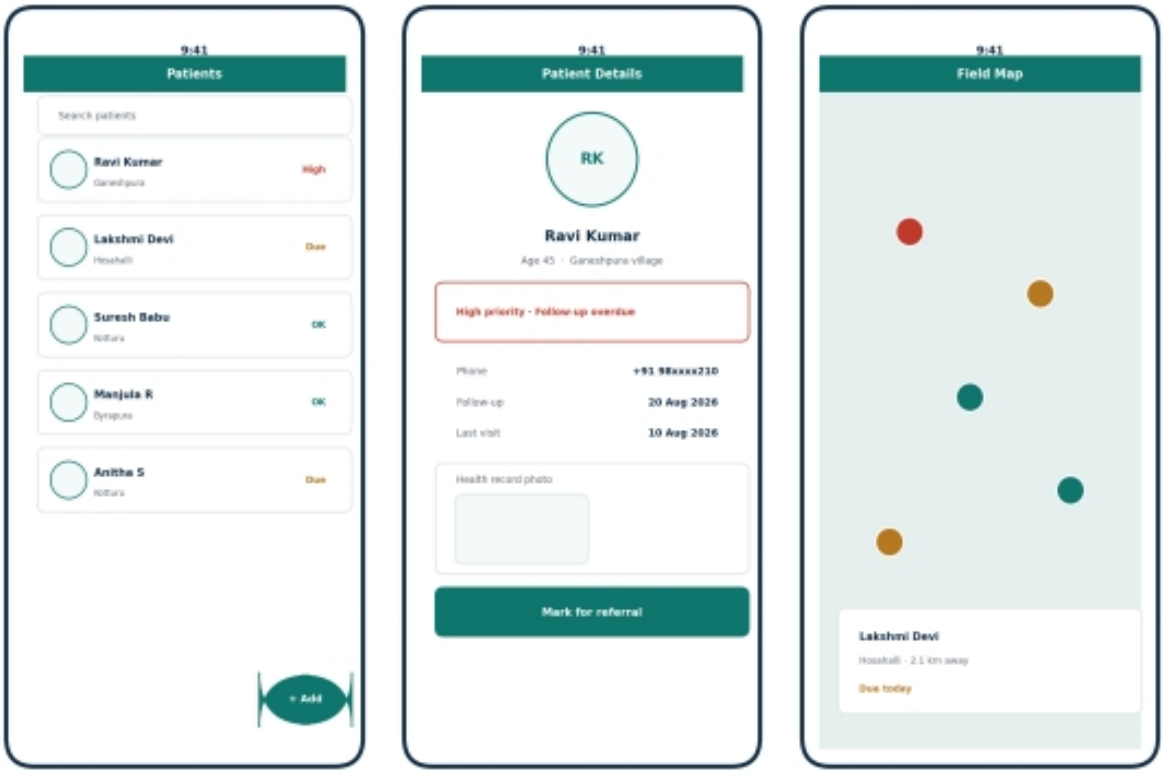
Splash screen Login Dashboard

Figure B.1 Entry and dashboard mockups

The splash screen establishes the application identity. Login provides the intended authenticated entry point. The dashboard gathers patient counts, priority groups and navigation to records, maps, reminders, reports and the assistant. Sample values are part of the visual design.

Review 3 verification: Confirm navigation, sign-in errors, session handling and accurate dashboard counts in the running build.

B.2 Patient records and field map



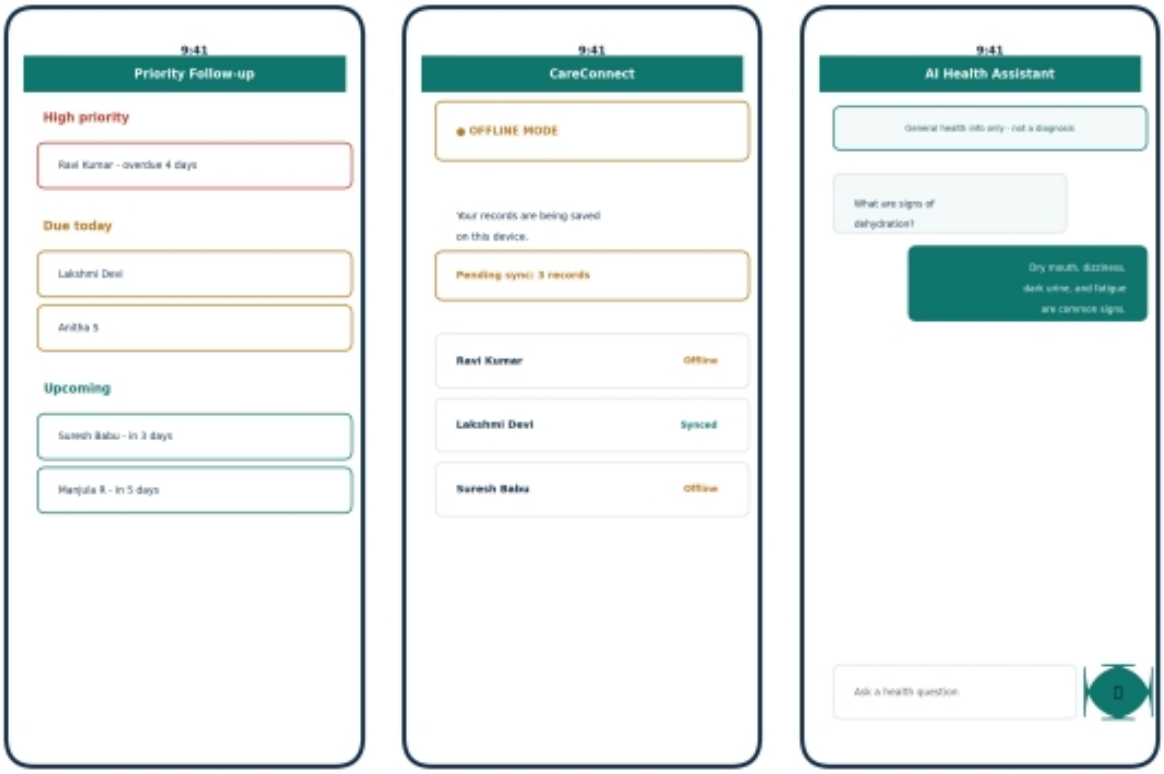
Patient management Patient details Field map

Figure B.2 Patient records and field map mockups

Patient management provides a searchable list and an add-patient action. Patient details groups information and follow-up status for one record. The field map shows example location markers. In implementation, marker selection should resolve to the same patient identifier used by the records module.

Review 3 verification: Confirm create/edit/search behavior, required-field validation and the correspondence between map markers and patient details.

B.3 Follow-up and assistance



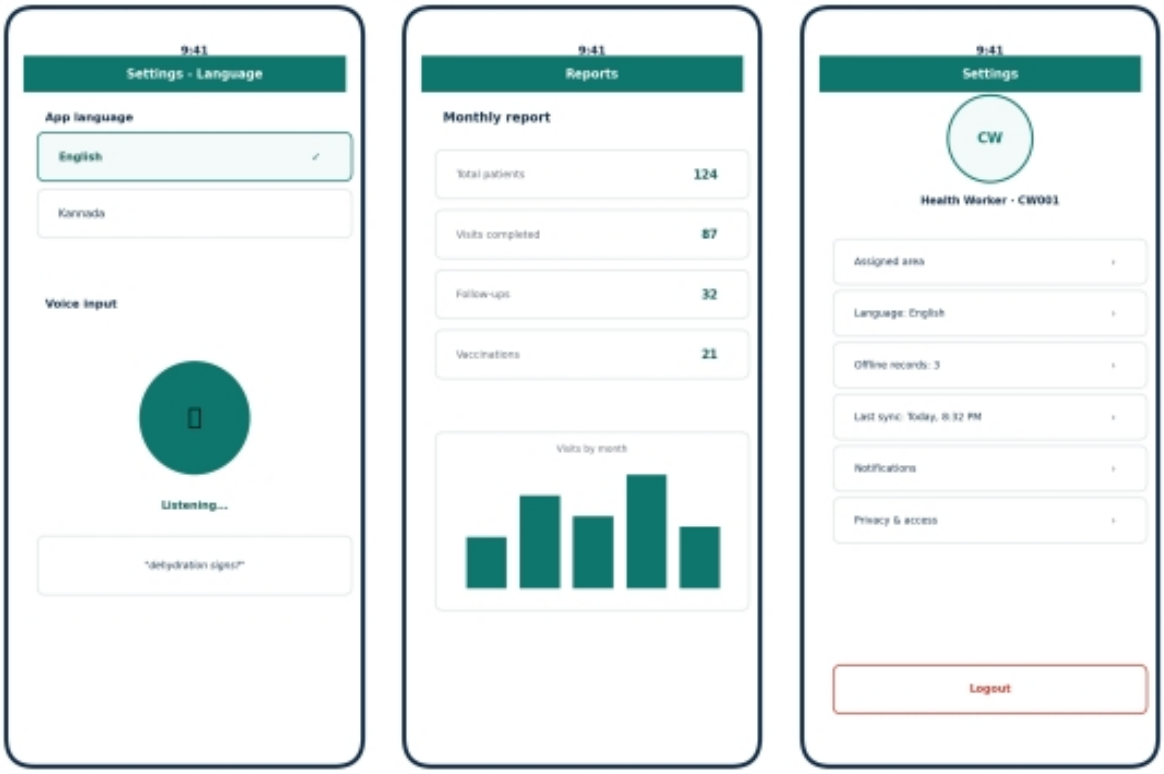
Priority follow-up Offline mode AI chatbot

Figure B.3 Follow-up and assistance mockups

The priority list separates High Priority, Due Today and Upcoming work. Offline mode communicates local availability and pending synchronization. The AI assistant mockup shows a conversational interface for general information; no response shown here should be treated as validated patient-specific guidance.

Review 3 verification: Confirm category boundary cases, offline persistence, visible sync errors and chatbot failure handling.

B.4 Language reports and settings



English/Kannada and voice Reports Settings

Figure B.4 Language reports and settings mockups

The language and voice screen illustrates English/Kannada selection and voice interaction. Reports illustrates operational counts and a chart; these are sample figures. Settings provides profile and application options, including language and offline/synchronization preferences.

Review 3 verification: Confirm readable translated labels, transcript correction, report calculations and persistence of saved preferences.

APPENDIX C ENCLOSURES AND SDG MAPPING

C.1 Enclosure status

Table C.1 records the documents available for this report. A research-paper manuscript is present, but it does not by itself establish publication, conference registration or acceptance. Missing certificates and reports have not been fabricated.

Table C.1 required enclosures and available evidence is presented below.

Table C.1 Required enclosures and available evidence

Required item	Status in the supplied material
Research paper	CareConnect manuscript supplied; its content is incorporated into this report.
Publication / conference certificate or registration proof	Not supplied. Include genuine proof for the students when available.
Project-related achievement or award certificate	Not supplied. No award claim is made.
Similarity / plagiarism report with percentage	Not supplied. No similarity percentage is stated.
SDG mapping	Included in Section C.2 below.

C.2 Mapping to Sustainable Development Goal 3

Table C.2 expands the project paper's SDG mapping. It describes intended support for health-service organization under Goal 3 (United Nations, n.d.), without claiming measured progress against an SDG indicator.

Table C.2 careConnect feature mapping to SDG 3 is presented below.

Table C.2 CareConnect feature mapping to SDG 3

Platform Function	SDG 3 Relevance
Patient tracking	Supports continuity of care.
Vaccination reminders	Support preventive-care activities.
Health information	Can improve access to general health information.
Medicine reminders	Support routine care activities.
Digital records	Improve organization of healthcare-service information.

Patient tracking and digital records support continuity of information across visits. Vaccination reminders support preventive-care scheduling, while medicine reminders support routine follow-up. General health-information access can help workers locate information, subject to appropriate review. Offline and language features are enabling design choices for field accessibility.

C.3 Project at a glance

Table C.3 careConnect project summary is presented below.

Table C.3 CareConnect project summary

Item	Details
Project Title	Community Health Worker Assistance Platform (CareConnect)
Category	Software
Program	B.Tech - Computer Science and Engineering
Platform	Android application
Core Database	Room + Firebase Firestore
Authentication	Firebase Authentication
AI Component	AI Chatbot API
Mapping	GIS / Field Map
Language	English and Kannada
Voice	Speech-to-text / voice interaction
Main Users	Community health workers
Key Theme	Offline-first field healthcare assistance
SDG Mapping	SDG 3 - Good Health and Well-being

Review stage: Review 3. Overall completion: 80% (team-reported). Project ID: PRJ-510. Earlier presentation panel: 12. The final submission should add actual acceptance results, runtime evidence and genuine required enclosures as they become available.